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30 June 1988
SUPERSEDING
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MILITARY SPECIFICATION

LATHES, WOODWORKING, PATTERNMAKERS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers patternmaking lathes for turning, shaping, and boring of wood and plastics, and for spinning nonferrous metal workpieces with equipment and accessories specified herein.

1.2 Classification. The specification covers lathes of the following types, classes, and sizes. The type, class, and size required shall be as specified (see 6.2.1).

Type I - Fixed bed (conventional) type.

Class A - Manually operated carriage
- Size 1 - 16 inches swing over bed - 60 inches between centers
- Size 2 - 20 inches swing over bed - 54 inches between centers
- Size 3 - 24 inches swing over bed - 72 inches between centers
- Size 4 - 30 inches swing over bed - 72 inches between centers

Class B - Manual and powered operated carriage.
- Size 1 - 16 inches swing over bed - 60 inches between centers
- Size 2 - 20 inches swing over bed - 54 inches between centers
- Size 3 - 24 inches swing over bed - 72 inches between centers
- Size 4 - 30 inches swing over bed - 72 inches between centers

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Type II - Bedless (modular) type.

Class A - Manually operated carriage
Size 1 - 92 inches swing over baseplate - 78 inches between centers
Size 2 - 96 inches swing over baseplate - 68 inches between centers
Size 3 - 92 inches swing, front and rear of headstock

Class B - Manual and power operated carriage
Size 1 - 92 inches swing over baseplate - 78 inches between centers
Size 2 - 96 inches swing over baseplate - 68 inches between centers
Size 3 - 92 inches swing, front, and rear of headstock

Type III - Gap type (sliding bed).

Class A - Manually operated carriage
Size 1 - 30 inches swing over bed - 78 inches between centers, with gap closed (see table III)
Size 2 - 30 inches swing over bed - 102 inches between centers, with gap closed (see table III)

Class B - Manual and power operated carriage
Size 1 - 30 inches swing over bed - 102 inches between centers, with gap closed (see table III)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specification and standards form a part of this specification to the extent specified herein. The issue of these documents shall be the latest revision of the document on the date of the solicitation and may include title changes.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

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MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interferences.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

2.1.2 Other Government documents and publications. The following other Government document forms a part of this specification to the extent specified herein. The issues shall be those in effect on the date of the solicitation and may include title changes.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issue of these documents shall be the latest revision of the document on the date of the solicitation and may include title changes.

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 360.02 - Manual for Machine Tool Gearing.

AGMA 390.03 - Gear Classification, Materials and Measuring Methods for Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717.)

AMERICAN NATIONAL STANDARD INSTITUTE (ANSI)

ANSI/NEMA MG 1 - Motors and Generators.

ANSI/B5.1 - T-Slots, Their Bolts, Nuts, and Tongues.

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ANSI/01.1 - Safety Requirements for Woodworking Machinery.

ANSI/ISO 54 - Cylindrical Gears For General Engineering and For Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

(Non-Government standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The lathe shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The lathe shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Design and construction dimensional systems. Unless otherwise specified (see 6.2.1), both the U.S. Customary System of Units (US) and the International System of Units (SI) may be used in the design and construction of the machine. In this specification, all measurements, dimensions, sizes, and capacities aregiven in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.2 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in US, SI, or dual (US and SI) units as specified by the procuring activity (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.2.1 Dials. All dials used to indicate machine axes position or tool movement shall be graduated in increments not larger than .01 inch. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a

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nonglare background with graduations that can be read from the operator's normal position. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.3 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the intended use, performance, and design life of the machine. The reclaimed materials shall be reprocessed or recycled in a manner which will restore them to the same chemical composition and physical properties as virgin materials.

3.2.4 Energy efficiency. The lathe and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency as governed by the developments available within the industry.

3.2.5 Controls. All operating controls shall be located convenient to the operator's work station(s).

3.2.6 Safety and health requirements. Covers, guards, or other safety devices shall be provided for the point of operation and all other parts of the lathe that present safety hazards. The safety devices shall not interfere with the operation of the lathe. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All lathe parts, components, mechanisms, and assemblies furnished on the lathe, whether or not specifically required herein, shall comply with all of the requirements of OSHA 29 CFR 1910 and ANSI 01.1. If a conflict arises between 29 CFR 1910 and ANSI 01.1, 29 CFR 1910 shall apply.

3.2.6 Mercury restriction. The lathe shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the lathe.

3.2.8 Environmental protection. Under the operating, service, transportation, and storage conditions described herein, the lathe shall not emit materials hazardous to the ecological system as prescribed by Federal, state, or local statutes in effect at the point of installation.

3.2.10 Lubrication. Means shall be provided to ensure lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable and shall have a low-lubrication level warning light that is in clear view of the machine operation normal workstation. Each lubricant reservoir shall have at least a 24-hour capacity and means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible without any disassembly of the machine.

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3.2.10 Interchangeability. All parts subject to replacement shall be manufactured to definite dimensions and tolerances. Replacement parts (except for hand-fitted parts such as ways and boxed bearings) shall be interchangeable without requiring modification. Replacement parts are those which are subject to wear or failure such as gears, bearings, shafts, feed screws, drive belts, pumps, and seals. Electrical replacement parts shall include, but not be limited to, motors, relays, fuses, switches, magnetics, pushbuttons, light bulbs, braking mechanisms, and solid-state devices. Tooling and accessories normally supplied as standard items shall also be considered replacement parts. Foreign-sourced replacement parts shall be interchangeable with domestically supplied components or be logistically supportable and readily available from a domestic source.

3.3 Construction. The lathe shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the lathe to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed to prevent loosening. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the lathe shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts, either US or SI, used on the lathe and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Gears. All gears, shafts, and pinions of spindle and axis drive train shall be designed and manufactured of a suitable ferrous material to meet or exceed the requirements of AGMA 360.02 and 390.03 for the English (US) system or

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ISO 54 for the metric (SI) system. The gears, shafts, and pinions shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears, shafts, and pinions in the drive train shall be hardened and ground steel. Hardness shall be not less than Rockwell C-48.

3.3.8 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem class and the emission and susceptibility requirements shall be as specified.

3.4 Components.

3.4.1 Base. The lathe base shall be either of box and pedestal shapes or combinations with baseplates having slots for components support and adjustment. Base mass and construction shall have strength and rigidity to sustain maximum operating capacity of lathe.

3.4.2 Bed. The lathe bed of type I and type III lathes shall be constructed to support the headstock, tailstock, and carriage assemblies and such other components mounted thereon. The bed of type II lathes shall be constructed to support the carriage assembly.

3.4.2.1 Type I. The type I lathe bed shall be constructed with parallel ways on the top to support and provide longitudinal adjustments of the carriage and tailstock. The swing capacity over the bed and between centers shall be not less than the requirements of table I.

3.4.2.2 Type II. Type II lathes shall have a modular type bedway supporting a carriage assembly. The bedway module shall consist of a pedestal mounted bedway supported by baseplates having T-slots running parallel to the axis of the headstock spindle. The T-slots shall allow the bedway pedestal to be adjusted to any longitudinal position. A different series of T-slots in the baseplates shall allow pedestal adjustment in a transverse direction to the axis of the headstock spindle. The pedestal shall allow the bedway to rotate 360 degrees, to position the ways parallel and perpendicular, or any other angular position to the spindle axis. The pedestal shall have a collar, graduated in 1 degree increments, indicated the angular position of the bedway to the spindle axis. The pedestal shall have means for securely clamping the bedway in any angular adjusted position.

3.4.2.2.1 Baseplates. The baseplates supporting the bedway shall be assembled to other baseplates which supports the pedestal mounted headstock and tailstock. All baseplates shall be standardized in construction so that swing capacity between centers is increased by additional baseplate assemblies. All baseplates shall have means for securely assembling all pedestal mounted components and baseplates together, means for leveling, and means for securely bolting the assembled lathe to the floor or foundation. Baseplate dimensions shall conform to the

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requirements of table II. Type II, size 3, lathes consist of a baseplate supporting a pedestal mounted headstock, baseplates supporting a pedestal mounted bedway and such other baseplates, necessary to form a lathe that can perform turning, facing and forming operations with the workpiece mounted on a faceplate. Type II, size 1 and 2, lathes shall consist of components required for size 3 lathes and the addition of a baseplate supporting a pedestal mounted tailstock, forming a lathe, that can also perform turning operations between centers. The baseplate supporting the tailstock and baseplates between the headstock and tailstock shall have T-slots that allow the tailstock to be adjusted to any location along the baseplates to vary the center-to-center distance. Means shall be provided for securely locking the tailstock pedestal in any position along its range of adjustment. Baseplate dimensions, swing capacity, and distance between centers shall be not less than the requirements of table II.

3.4.2.3 Type III. The bed of type III lathes shall be constructed in two sections with the top section fitted on matched ways of the lower, combination, base and bed section. The outer end of the upper section shall be supported by an adjustable leg on casters. The upper section of the bed shall have parallel ways to provide longitudinal movements and adjustments of the tailstock and carriage. The wing over the bed, through the gap and between centers shall be not less than the requirements of table III.

3.4.3 Headstock. The headstock shall be constructed to support the spindle, workpiece mounting and driving devices, and workpieces mounted thereon. The spindle and spindle speed components attached to the spindle, shall be supported by the headstock on ball or roller bearings of a type, class and size suitable for the application. The headstock shall be fitted with a removable cover or guard over the spindle drive assembly, allowing access to the cover parts.

3.4.3.1 Spindle. The spindle shall be hollow to provide for removal of spindle centers. Spindles shall be constructed for mounting of faceplates on both ends. The spindle end, facing the tailstock, shall have a Number 4 Morse taper. The spindle shall be supported by antifriction bearings of a type, class and size suitable for the application. Means shall be provided for spindle bearing adjustment.

3.4.3.2 Spindle brake and lock. Type I lathes shall have a manually operated brake and spindle locking mechanism. The brake shall stop spindle rotation at its greatest rpm within a time period of 15 seconds after brake application. The spindle locking mechanism shall prevent spindle rotation during faceplate installation and removal. Controls for the spindle brake and locking mechanism shall be conveniently accessible on the operator's side of the headstock.

3.4.4 Tailstock. A tailstock shall be furnished on all lathes except type II, size 3. The tailstock shall be mounted to support the free end of the workpiece. The position of the tailstock shall be adjustable to accommodate workpieces having

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any length up to the rated center-to-center dimension. The tailstock shall incorporate a set-over feature to facilitate taper turning of workpieces. The set-over range shall be not less than 1 inch in each direction off center. The set-over range shall be marked on a scale, on the base of the tailstock, in increments not greater than 1/8 of an inch.

3.4.4.1 Tailstock spindle. The tailstock spindle shall be horizontally adjustable, by means of a handwheel operated screw, for a distance not less than 6 inches. The tailstock spindle shall receive the shank of centers or other adapters having a Number 4 Morse taper and shall provide a means for tool ejection. The tailstock spindle shall have a locking device which retains the spindle in horizontal adjustment while supporting the free end of a workpiece.

3.4.5 Carriage. All lathes shall have a carriage mounted to provide longitudinal movements and support the cutting tools mounted in a compound tool holder. The class A lathe carriage shall have a manually operated mechanism that feeds the carriage longitudinally with the spindle axis. The class B lathe carriage shall have both manual and powered longitudinal feed mechanisms for cutting tools mounted in the tool holder. All carriages shall have means for securely clamping the carriage to their bedways to prevent longitudinal movement during facing operations.

3.4.5.1 Carriage drive. Class A lathe carriage movements shall be accomplished by means of manually driven gearing mounted within the carriage apron. A handwheel shall provide the manual means of the carriage drive. Carriage movements of the class B lathes shall be both manual and power driven. Unless otherwise specified (see 6.2.1), powered drives for class B lathes shall be independent, infinitely variable speed carriage drive. The independent drive shall provide forward, neutral, and reverse controls. The infinite speed range of the carriage shall be from 0 to not less than 50 inches per minute. Apron gearing on class B lathes shall provide means of selecting either manual or powered carriage drive. The powered feed mechanism shall include either a shear device, a slip clutch or other means to prevent damage to the feed mechanism if the carriage is fed against a stationary object.

3.4.5.2 Cross slide. All lathes shall have a manually operated cross slide supported on the carriage. The cross slide shall feed the cutting tool in a direction perpendicular to the axis of the spindle. The cross slide and its mating ways shall have adjustable gibs. The cross slide shall have a manually operated crank or handwheel that operates a feedscrew that traverses the cross slide. The crank or handwheel collar, or the cross slide itself, shall have graduations indicating the cross slide movement. The graduations shall be in increments not larger than 1/8 inch.

3.4.5.3 Compound slide. All lathes shall have a compound slide mounted on the cross slide. The compound slide shall swivel through 360 degrees and shall have a scale graduated in increments not larger than 1 degree to indicate the angle to

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which the slide is swiveled. Means shall be provided to securely clamp the slide in any position within its range of rotation. The compound slide shall have a manually operated crank or handwheel that operates a feedscrew that feeds the compound slide. The crank or handwheel collar or the slide itself shall have graduations indicating cross slide movement in increments not larger than 1/8 inch. The compound slide shall have a T-slot conforming to ANSI B5.1 that will accommodate a tool post. The compound slide shall be detachable so that a hand tool rest may be installed.

3.4..6 Electrical system. The electrical system shall conform to ANSI/NFPA 79. Unless otherwise specified (see 6.2.1), each machine shall draw all of its electrical power from a single 230/460-volt, 3-phase, 60-hertz (Hz) circuit having a fused safety disconnect switch or circuit breaker. The machine shall be initially wired for operation on 460 volts.

3.4.6.1 Motors, type I lathes. Unless otherwise specified (see 6.2.1), the rotor shaft of the motor for type I lathes shall be integral with the headstock spindle. When specified (see 6.2.1), the motor shall drive the headstock spindle either by a four-speed, eight-speed, or alternating-current/direct current (DC) infinitely variable speeds, as specified (see 3.4.7.1).

3.4.6.2 Motor, type II and III lathes. Unless otherwise specified (see 6.2.1), the motor on type II and III lathes shall be mounted on a hinged base to facilitate spindle speed changes through the countershaft and maintain belt tension.

3.4.7 Drive.

3.4.7.1 Type I lathes. Unless otherwise specified (see 6.2.1), the headstock spindle for type I lathes shall be directly driven by the motor. The drive shall provide spindle speeds conforming to the requirements of table I. A manually operated spindle speed selector shall be located on the operator's side of the headstock. Spindle speeds shall be permanently and legibly marked corresponding to each of the four positions of the spindle speed selector. When specified (see 6.2.1), type I lathes shall be furnished with one of the optional spindle drives, A through I, listed in table I (see 3.4.6.1).

3.4.7.2 Type II and III lathes. The headstock spindle for type II and III lathes shall be belt driven by step pulleys through a countershaft. Belt tension shall be maintained and spindle speed changes manually available from the operator's side of the headstock. Unless otherwise specified (see 6.2.1), spindle speeds shall conform to tables II and III.

3.4.8 Hour meter. When specified (see 6.2.1), the lathe shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 9,999 hours in increments of not greater than 1 hour.

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The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the lathe.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the lathe shall be not less than specified in table I for the machine size specified.

3.6 Performance. When tested in accordance with the requirements of 4.8.2 through 4.8.3, the machine shall be capable of meeting the following performance accuracies:

Facing (workpiece mounted on faceplate):
Faced surfaces shall be flat within 0.005 inch per foot of workpiece diameter.

Between centers turning:
Out-of-round - Not to exceed 0.003 inch.
Taper - Not to exceed 0.003 inch per foot.

3.7.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.7 Alignment tolerances. Each lathe shall meet the alignment tolerances specified in table IV.

3.8 Standard equipment. Unless otherwise specified (see 6.2.1), standard equipment shall consist essentially of one set of wrenches, tool rests, faceplates, cutters, and cutter holder normally furnished with the lathe by the supplier.

3.9 Optional equipment. Optional equipment shall be furnished as specified (see 6.2.1).

3.10 Marking on plates. All words on plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. All plates shall be corrosion-resistant.

3.10.1 Lubrication plate. A lubrication plate shall be permanently and securely attached to each machine. Unless otherwise specified (see 6.2.1), the plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity
Military or Federal specification number for each lubricant
Size and type of each lubricant filter

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3.10.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2.1), the nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model number of the basic standard machine and a suffix identified in the manufacturer's permanent records.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Technical data. When technical data is required, it shall be furnished in accordance with the requirements of DD Form 1423. All technical data furnished shall be written in the English language.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements. Unless otherwise specified in the contract or purchase order, all inspections shall be performed within the limits of the continental U.S.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

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a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.3).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all inspections, tests, and examinations shall be performed in an indoor facility with ambient conditions of 55°F to 100°F and 10 percent to 95 percent relative humidity.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. Failure of the item to pass the first article examination or any test shall be cause for rejection.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.7, the tests in 4.8, and the inspection in 4.3. Failure of the item to pass the examination, the test, or the inspection shall be cause for rejection.

4.6 Acceptance test. Acceptance test shall be performed on the lathe to ensure conformance with this specification. Unless otherwise specified (see 6.2.1), the acceptance test shall be performed only after the machine is installed at its final location. The acceptance test shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.7 Examination. The lathe shall be examined to determine compliance with all requirements of this specification.

4.8 Tests.

4.8.1 Operational test. The lathe shall be operated at no load for not less than 15 minutes. Proper operation of all controls, motors, and adjusting mechanisms and other accessories shall be verified during trial period.

4.8.2 Performance tests. The material of the workpieces shall be at the supplier's option. Workpieces made of seasoned hardwood, plastic, soft nonferrous metal, or any similar material, provided that the material characteristics will allow measurements to the specified accuracies. The size of the workpiece used for the facing test shall be such that the diameter of the faced surface will be not smaller than the largest faceplate supplied with the lathe. The surface of the workpiece contacting the faceplate shall be finished flat prior to mounting to the face plate. The method of attachment to the faceplate, for the facing test, and the method of driving the workpiece, for the between centers test, shall be at the option of the supplier.

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4.8.2.1 Facing test. All lathes shall perform the following facing test. A workpiece conforming to the requirements of 4.5.3 shall be securely attached to the faceplate. One rough facing cut and one finish facing cut shall be taken across the face of the workpiece. The type of cutting tool used shall be at the option of the supplier. The depth of the rough cut shall be at the option of the suppler, but the depth of the finish cut shall be not less than 0.005 inch. The faced surface shall be measured for flatness. The faced surface shall be flat within the tolerances specified in 3.6.

4.8.2.2 Between centers turning test. All lathes, except type II, size 3, shall perform the following turning test. The test shall be conducted on a workpiece conforming to the requirements of 4.8.2 of a size that will produce a cylinder not less than 3 inches outside diameter and of a length not shorter than half the rated distance between centers. The workpiece shall be prepared for mounting between centers and shall be mounted in that manner. If the workpiece is square, it shall be roughly turned to cylindrical form. One rough longitudinal and one finish longitudinal cut shall be taken along the length of the workpiece. The type of cutting tool used shall be at the option of the supplier. The depth of the rough cut shall be at the option of the supplier, but the depth of the rough cut shall be the option of the supplier, but the depth of the finish cut shall be less than 0.005 inch. The finished surface shall be measured for roundness and straightness. The finished cylinder shall be round and straight within the tolerances specified in 3.6.

4.8.2 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The lathes covered by this specification are intended for use in patternmaking shops where patternmaking operations are performed.

6.2 Ordering data.

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6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, and size required (see 1.2).

c. First article, if required (see 3.1).

d. If lathe is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. Specify system of units for measuring and indicating devices (see 3.2.2).

f. Painting, if different (see 3.3.5).

g. Electromagnetic interference control, if required (see 3.3.8).

h. Carriage drive, if different (see 3.4.5.1).

i. Electrical system, if different (see 3.4.6).

j. Motor enclosure, if different (see 3.4.6.1).

k. Drive, if different, specify option indicating the specific motor h.p., speed range required (see 3.4.7.1 and 3.4.7.2).

l. Standard equipment, if different (see 3.8).

m. Optional equipment, if required (see 3.9).

n. Lubrication chart, if different (see 3.10.1).

o. Nameplate, if different (see 3.10.2).

p. First article inspection, if different (see 4.4).

q. Quality conformance inspection, if different (see 4.5).

r. Preservation, packing and marking if different (see 5.1).

s. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level required (see 5.1).

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6.2.2 Consideration of data requirements. The requirements for technical data should be considered when this specification is applied on a contract. The Data Item Descriptions (DIDs), DD Form 1664, for required data should be listed on a Contract Data Requirements List (DD Form 1423). The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements List, (AMSDL) should be researched for the applicable DD Form 1664 and to ensure that only current approved DIDs are cited on the DD Form 1423.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.5 Training. Training required by the procuring activity shall be provided as specified in the contract.

6.6 Warranty. Warranty requirements shall be as specified by the procuring activity in the contract.

6.7 Subject term (keyword) listing.

Lathe
Patternmaking
Woodworking

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TABLE I. Sizes and capacities.

Type I, classes A and B, fixed bed, manual and power feed carriage.

Characteristics	Class A and B			
	Size 1	Size 2	Size 3	Size 4
Swing, inches				
Over bedway	16	20	24	30
Over cross slide	13	16	19	25
Distance between centers, inches	60	54	72	72
Tool holder size, inches	3/4 x 2-1/8			
Cross slide traverse distance, inches	15	15	17-1/4	15-1/4
Compound slide traverse distance, inches	5	5	7	7
Faceplates to be furnished, dia., inches	8, 12, 20 and 24			
Spindle drive system:				
(The table enteries are in the following order: Spindle motor HP - Number of spindle speed selections - spindle speed range in RPM. See Example below*)				
Standard drive system	2-4-600 to 3600			
Optional drive systems:				
Option A	2-4-360 to 2700			
Option B	2-4-600 o 1800			
Option C	3-4-600 to 1800			
Option D	3-4-600 to 1800			
Option E	5-4-350 to 2500			
Option F	3/1-1/2-8-180 to 2700			
Option G	5/2-1/2-8-170 to 2500			
Option H	3-inf. var. - 85 to 3000			
Option I	5-inf. var. - 85 to 3000			

* Example: 2-4-600 to 3600 means "2 Hp drive motor, 4 spindle speeds 600 RPM or less to 3600 RPM, or more, spindle speed range.

NOTE: Requirements are not less than those shown. Where a range is shown, the required performance is from the smaller figure or less to the larger figure or more.

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TABLE II. Sizes and capacities.

Type II, classes A and B
Bedless (modular) type, manually and power feed carriage

Characteristics	Size 1	Size 2	Size 3
Swing, inches:			
Spindle, Front, over baseplate	92	96	92
Spindle, Rear, over floor	100	104	96
Distance between centers, inches	78	68	None
Number of spindle speed selections	16		
Spindle speed range, RPM	75 thru 1500	35 thru 1350	75 thru 1500
Pedestal mounted bedway, dimensions, in.			
Length of bedway	57	57	57
Carriage travel on bedway	44	44	44
Tool holder size, inches	3/4 X 2-1/8		
Longitudinal baseplate size, inches			
Length, total	144	144	112
Width	32	32	32
Thickness	4	4	4
Bedway baseplate size, inches			
Length	64		
Width	48		
Thickness	4		
Motor horsepower	10/5 (2 speed)		
Cross slide traverse distance, inches	16-1/4		
Compound slide traverse distance inches	7		
Faceplate to be furnished dia., inches	12, 24, 30 and 38		

NOTE: Requirements are not less than those shown. Where a range is shown, the required performance is from the smaller figure or less to the larger figure or more.

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TABLE III. Sizes and capacities.

Type III, classes A and B
Gap type, manual and power feed carriage.

Characteristics	Class A		Class B
	Size 1	Size 2	Size 1
Swing, inches:			
Gap closed:			
Over bedway	30		
Over carriage	26		
Gap open:			
Swing thru gap	48	60	60
Width of gap	24	92	92
Distance between centers, inches:			
Gap closed	78	102	102
Gap open	102	192	192
Carriage travel distance, inches	78	105	105
Tool holder size, inches	3/4 X 2½	3/4 X 1-3/4	3/4 X 1-3/4
Number of spindle speed selections	16		
Spindle speed range, RPM	90 thru 1750	75 thru 1500	75 thru 1500
Motor horsepower	10/5 (2 speed)		
Cross slide traverse distance, inches	13	14	14
Compound slide traverse distance, inches	7	9	9
Faceplates to be furnished, dia., inches	12, 24, 30, and 38		

NOTE: Requirements are not less than those shown. Where a range is shown, the required performance is from the smaller figure or less to the larger figure or more.

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TABLE IV. Alignment tolerances.

Test	Permissible error	
	Types I and III	Type II
Center runout (center installed in spindle taper; measured perpendicular to taper at center of taper	0.002 inch, TIR	
Spindle internal taper runout (measured on precision test bar inserted in spindle taper; measured 1 inch from spindle end)	0.002 inch, TIR	
Faceplate runout, axial (measured on face of faceplate, at outer diameter)	0.002 inch per foot	
Spindle axis parallel to bedways (measured in both vertical and horizontal planes)	0.003 inch per foot	0.005 inch per foot

NOTE: Cross slide travel squareness with spindle axis and alignment of head and tail centers to be tested by turning a workpiece.

Custodians:
Navy - SH
Air Force - 99

Review activities:
Navy - SH
Air Force - 84

User activities:
Navy - MC

Preparing activity:
DLA - IP

Project (3220-0152)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-80047D	Lathes, Woodworking, Patternmakers

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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